

# HOIN JUNG

Purdue University, West Lafayette, IN, USA

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## EDUCATION

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| <b>Purdue University</b><br><i>Ph.D. in Electrical and Computer Engineering</i><br>· Dissertation: Efficient Surgical Interventions for Trustworthy Multimodal Machine Learning Systems | <b>West Lafayette, IN, USA</b><br>Jan. 2023 – Present |
| <b>Seoul National University</b><br><i>M.S. in Computational Science and Technology</i><br>· Thesis: Local-Ensemble Graph Collaborative Filtering with Spectral Co-Clustering           | <b>Seoul, Korea</b><br>Sep. 2020 – Aug. 2022          |
| <b>Korea Aerospace University</b><br><i>B.E. in Aerospace &amp; Mechanical Engineering</i><br>· Major in Aircraft System Engineering                                                    | <b>Goyang, Korea</b><br>Mar. 2010 – Feb. 2014         |

## RESEARCH INTERESTS

### Trustworthy & Efficient AI Systems

#### Research Context

- Primary Domains: Multimodal Learning, Retrieval-Augmented Generation, Scientific and Engineering AI
- Key Problems: Factual Grounding, Fairness & Debiasing, Uncertainty, Distribution Shift

#### Research Methodology

- Primary Approach: Efficient Post-Hoc and Inference-Time Interventions
- Key Techniques: Causal Discovery, Latent Feature & Logit Adjustment, Multimodal Explainability

## AWARDS AND SCHOLARSHIPS

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| <b>Bilsland Dissertation Fellowship</b><br><i>Purdue University</i><br>• Awarded to outstanding Ph.D. candidates to support final dissertation completion.                                                                                                             | Spring 2027             |
| <b>KDD 2026 Research Track Best Reviewer (370 selected)</b>                                                                                                                                                                                                            | Aug. 2026               |
| <b>CVPR 2026 Outstanding Reviewer (Top 5%)</b>                                                                                                                                                                                                                         | Jun. 2026               |
| <b>Advancing Machine Learning for Scientific Discovery Fellowship</b><br><i>Los Alamos National Laboratory</i><br>• Awarded to select graduate researchers to develop trustworthy and transparent machine learning methodologies for critical scientific applications. | Summer 2026             |
| <b>Purdue University Nominee for Google PhD Fellowship (4 of 32 selected)</b>                                                                                                                                                                                          | Apr. 2026               |
| <b>CVPR 2025 Outstanding Reviewer (Top 5%)</b>                                                                                                                                                                                                                         | Jun. 2025               |
| <b>Purdue Graduate Student Government Travel Grant</b>                                                                                                                                                                                                                 | Nov. 2024               |
| <b>NeurIPS 2024 Scholar Award</b>                                                                                                                                                                                                                                      | Oct. 2024               |
| <b>NeurIPS 2024 Spotlight Paper (Top 2.5%)</b>                                                                                                                                                                                                                         | Oct. 2024               |
| <b>Future Industry Talent Graduate Scholarship</b><br><i>Hyundai Motor Chung Mong-Koo Foundation</i>                                                                                                                                                                   | Fall 2021 – Spring 2022 |
| <b>National S&amp;T (Science &amp; Technology) Scholarship</b><br><i>Korea Student Aid Foundation</i>                                                                                                                                                                  | Fall 2010               |

## PUBLICATIONS

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1. **H.Jung** and X.Wang, “Uncertainty-Aware Hybrid Retrieval for Long-Document RAG”, *Empirical Methods in Natural Language Processing (EMNLP)*, Main Conference, Long Paper, 2026. (Accepted)
2. **H.Jung** and X.Wang, “Beyond Chunking: Efficient Global Pooling for Holistic Long-Document Representation”, *Empirical Methods in Natural Language Processing (EMNLP)*, Main Conference, Long Paper, 2026. (Accepted)
3. **H.Jung**, S.Lu, D.Wang, and X.Wang, “Assessing the Reliability of Image Quality Metrics and Mitigating Quality Bias in Generative Models”, *Conference on Computer Vision and Pattern Recognition (CVPR)*, Findings, pp. 7956–7967, 2026.
4. **H.Jung**, V.Nascimento, H.Liu, X.Wang, C.K.Koh, and D.Jiao, “Generative Antenna Design with Accuracy, Diversity, and Explainability via Dynamic Pseudo-Labeling”, *IEEE Journal on Multiscale and Multiphysics Computational Techniques (JMMCT)*, vol. 11, pp. 246–257, 2026.
5. **H.Jung**, J.Chai, and X.Wang, “Adaptive Logit Adjustment for Debiasing Multimodal Language Models”, *International Conference on Learning Representations (ICLR)*, pp. 53666–53692, 2026.
6. **H.Jung**, J.Liu, A.Rao, H.Kim, X.Zhao, A.Chandra, and M.Sarkis, “TVAgent: A lightweight Vision-Language-Model for TV GUI Agent”, *AAAI Workshop on Deployable AI (DAI)*, 2026.
7. T.Jang, **H.Jung**, and X.Wang, “Target Bias Is All You Need: Zero-Shot Debiasing of Vision-Language Models with Bias Corpus”, *International Conference on Computer Vision (ICCV)*, pp. 1935–1946, 2025.
8. **H.Jung**, J.Chai, and X.Wang, “Adversarial Latent Feature Augmentation for Fairness”, *International Conference on Learning Representations (ICLR)*, pp. 61305–61340, 2025.
9. H.Lee, **H.Jung**, and S.Bae, “Framing Korea: the role of international student YouTubers in shaping destination perceptions”, *Current Issues in Tourism*, pp. 1–16, 2025.
10. **H.Jung** and X.Wang, “Towards On-the-Fly Novel Category Discovery in Dynamic Long-Tailed Distributions”, *Winter Conference on Applications of Computer Vision (WACV)*, pp. 6795–6804, 2025.
11. **H.Jung** and X.Wang, “Fairness-Aware Online Positive-Unlabeled Learning”, *Empirical Methods in Natural Language Processing (EMNLP)*, Industry Track, pp. 170–185, 2024.
12. **H.Jung**, T.Jang, and X.Wang, “A Unified Debiasing Approach for Vision-Language Models across Modalities and Tasks”, *Neural Information Processing Systems (NeurIPS)*, pp. 21034–21058, 2024. (**Spotlight**)
13. **H.Jung**, V.Nascimento, H.Liu, X.Wang, C.K.Koh, and D.Jiao, “Explainable Planar Multiband Antenna Designer with Wasserstein Generative Adversarial Network”, *IEEE International Symposium on Antennas and Propagation (AP-S)*, 2024. (**Oral Presentation**)
14. **H.Jung**, H.S.Choi, and M.Kang, “Boundary Enhancement Semantic Segmentation for Building Extraction From Remote Sensed Image”, *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 60, pp. 1–12, 2022.

## PAPERS UNDER REVIEW AND MANUSCRIPTS

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1. **H.Jung** and X.Wang, “The Cost of Context: Mitigating Textual Bias in Multimodal Retrieval-Augmented Generation”, *Neural Information Processing Systems (NeurIPS)*, 2026.
2. **H.Jung**, D.DeSantis, A.Murph, J. Strait, and X.Wang, “CausalDrift: A Large-Scale Benchmark for Causal Discovery Under Temporal Drift”, *Transactions on Machine Learning Research (TMLR)*.
3. **H.Jung**, H.Liu, K.Chao, X.Wang, C.K.Koh, and D.Jiao, “Calibrated Uncertainty-Aware Graph Surrogates for Broadband S-Parameter Estimation of Chiplet Interconnects”, *IEEE Transactions on Components, Packaging and Manufacturing Technology (T-CPMT)*.
4. S.Lu, **H.Jung**, Z.Fang, and X.Wang, “Open-World Diversity Guidance for Fair Image Generation,” *Winter Confer-*

ence on Applications of Computer Vision (WACV), 2027.

5. X.Liu, R.Zhang, W.Yu, S.Xiong, H.Liu, F.Wu, **H.Jung**, M.Fredrikson, X.Wang, and J.Gao, "The Vision Wormhole: Latent-space Communication in Heterogeneous Multi-Agent System", *International Conference on Learning Representations (ICLR)*, 2027.
6. C.Han, Y.Sim, **H.Jung**, J.Lee, H.Lee, YS.Kang, S.Woo, G.Kim, HW.Park, and M.Jun, "IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer", *Manuscript*, 2026.

## TEACHING EXPERIENCE

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### ECE 695 Machine Learning in Bioinformatics and Healthcare

Guest Lecture

Purdue University, West Lafayette, IN

Fall 2025

- Designed and delivered a lecture on "Post-Training and Inference-Time Approaches for ML in Healthcare," introducing advanced efficiency techniques to graduate students.

### ECE 570 Artificial Intelligence

Teaching Assistant, Electrical & Computer Engineering

Purdue University, West Lafayette, IN

Fall 2024, Spring 2025, Spring 2026

- Held office hours and led student projects across three course sections, serving a total of 804 students.

### Computer Literacy & Programming (Python)

Instructor, Language Education Institute

Seoul National University, Seoul, Korea

Mar. 2021 – Jul. 2022

- Designed and delivered a Python programming course for beginner-level students for three semesters.
- Earned an average course evaluation score of **4.91/5.00** across 14 sections.

### L0444: Basic Computing (Python)

Teaching Assistant, Faculty of Liberal Education

Seoul National University, Seoul, Korea

Spring 2021, Spring 2022

- Led weekly lab sessions for 50+ students each semester.

## WORK EXPERIENCE

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### Los Alamos National Laboratory (LANL)

Advancing Machine Learning for Scientific Discovery Fellowship (Graduate Internship)

Los Alamos, NM, USA

May 2026 – July 2026

- Developed a computational benchmark for evaluating causal-discovery methods on nonlinear, nonstationary time-series data with changing causal structures.
- Implemented and evaluated machine-learning and statistical causal-discovery baselines in Python using standardized windowing protocols and graph-recovery metrics.
- Designed experiments to analyze the effects of causal drift, observation-window length, and model assumptions on causal-graph identification performance.

### Rapid Heterogeneous Integration Design Institute

Research Assistant, Elmore ECE Emerging Frontiers Center

West Lafayette, IN, USA

Jan. 2023 – Present

- Developed a generative framework for multiband planar antenna design.
- Used SHAP values to explain predictions from machine-learning-based electromagnetic models.
- Developed calibrated uncertainty-aware graph surrogates for broadband S-parameter estimation of chiplet interconnects.

### Samsung Research America

Research Scientist Intern

Irvine, CA, USA

May 2025 – Aug. 2025

- Conducted research on AI-driven Smart TV solutions in collaboration with senior engineers.
- Developed an automatic keyboard navigation system using vision-language models (VLMs), specifically optimized for commercial deployment.

### Samsung Electronics

Engineer, R&D Team, Department of Digital Appliance

Suwon, Korea

Aug. 2017 – Aug. 2020

- Designed and evaluated thermal-fluid systems for new refrigerator models.
- Analyzed and optimized refrigeration-cycle controls to reduce power consumption.

#### **ROK Air Force**

*Lieutenant, Aircraft Maintenance Officer, the 19<sup>th</sup> Fighter Wing*

**Chungju, Korea**

Jun. 2014 – May 2017

- Managed aircraft line maintenance and administered ground safety department for the military base.

### **ACADEMIC SERVICE**

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#### **Conference Reviewing & Program Committees**

- Area Chair: IEEE ICASSP (2026)
- Program Committee: AAAI (2024, 2025, 2026), IEEE BigData (2025, 2026)
- Reviewer: ICLR (2026, 2027), CVPR (2025, 2026), WACV (2026, 2027), NeurIPS (2025, 2026), SafeMM-AI Workshop (ICCV 2025), ECCV (2024), KDD (2024, 2026)

#### **Journal Reviewing**

- IEEE Journal on Multiscale and Multiphysics Computational Techniques (JMMCT)
- Transactions on Machine Learning Research (TMLR)
- IEEE Transactions on Geoscience and Remote Sensing (TGRS)

#### **University Service**

- Grant Review and Allocation Committee, Purdue Graduate Student Government

### **LEADERSHIP**

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**Mentor**, Purdue ECE G-LaMP (Graduate Leadership and Mentorship Program) (2025-2026)

**Co-Chair**, ICON Student Research Conference, Purdue University (2026)

**Vice President**, Student Government, Korea Aerospace University (2013)

### **SELECTED PRESENTATIONS**

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1. “Generative Antenna Design with Accuracy, Diversity, and Explainability via Dynamic Pseudo-Labeling”  
*Journal Seminar (Virtual), IEEE JMMCT* Jul. 2026
2. “Assessing the Reliability of Image Quality Metrics and Mitigating Quality Bias in Generative Models”  
*Poster, CVPR 2026 Findings* Jun. 2026
3. “Adaptive Logit Adjustment for Debiasing Multimodal Language Models”  
*Poster, Purdue ECE Open House Symposium* Mar. 2025
4. “A Unified Debiasing Approach for Vision-Language Models across Modalities and Tasks”  
*Spotlight Poster, Neural Information Processing Systems (NeurIPS 2024)* Dec. 2024
5. “An Efficient and Unified Debiasing Approach for Vision-Language Models across Modalities and Tasks”  
*Lightning Talk, Fast Machine Learning for Science Conference 2024* Jul. 2024
6. “Explainable Planar Multiband Antenna Designer with Wasserstein Generative Adversarial Network”  
*Oral Presentation, 2024 IEEE International Symposium on Antennas and Propagation* Jul. 2024
7. “Boundary Improvement Module for Binary Semantic Segmentation in Remote Sensing”  
*Oral, Korean Society for Industrial and Applied Mathematics (KSIAM)* Jun. 2021
8. “Segmentation model for tracking building in satellite imagery”  
*Poster, Korean Society for Industrial and Applied Mathematics (KSIAM)* Nov. 2020